TAKE-HOME WALKTHROUGH

Raw CSVs become CDM-ready data



The implementation covers all four requirements: transform, store, describe and diagram.

5	contacts
4	valid orders
1	orphan rejected
ISO	UTC timestamps

Result: clean entity CSVs, a validated model.json and traceable rejected rows.

Four steps cover the brief

Each requirement maps to one clear part of the implementation.

1 ✓

Parse & map

Read irregular CSV rows, clean fields and validate entity models.

src/cleaning.py · src/etl.py

→

2 ✓

Store

Write one CSV per CDM entity in the required folder structure.

cdm_lakehouse/

→

3 ✓

Describe

Generate model.json from the same entity definitions used for CSV headers.

src/manifest.py

→

4 ✓

Explain

Document field mappings, 1:N relation and storage layout with Mermaid.

README.md

AUTO pipeline execution

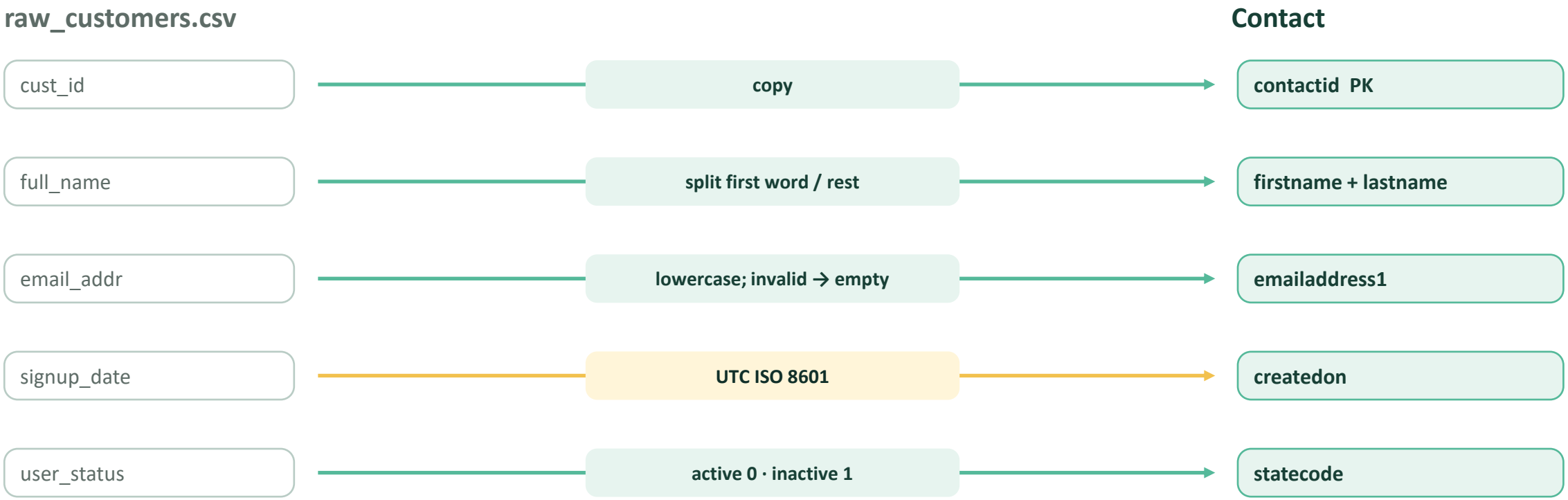
MODEL Pydantic + CDM metadata

PEOPLE logs, quarantine, README

How: python -m src.main

Contact: six mappings, three cleaning rules

The target row is validated before it reaches Contact.csv.



✓ CST-005 split line is joined; Alice's invalid email is logged and left empty.

SalesOrder: keep only valid relationships

Contact is the parent; SalesOrder.customerid is the foreign key.

PARENT

Contact

contactid PK

firstname

lastname

emailaddress1

createdon · statecode

1 — 0..N

contactid <- customerid

ORD-1005 · CST-999

No matching Contact → quarantine + log

CHILD

SalesOrder

salesorderid PK

customerid FK

totalamount · decimal

createdon · UTC

Exactly one parent is required for every retained order.

DELIVERED

The output is self-describing

model.json links the entity definitions, partitions and relationship.

cdm_lakehouse/

↳ **model.json**	entities · attributes · 1:N relationship
↳ **Contact/Contact.csv**	5 clean contacts
↳ **SalesOrder/SalesOrder.csv**	4 valid orders

Evidence

46	tests passed
97%	coverage
2	Python versions
3	README diagrams

- ✓ CI green on Python 3.11 and 3.13
- ✓ Manifest checked against the supplied reference and schema

Brief complete: data, metadata, relationships and documentation.